

EduTracker ERP: Comprehensive Client Deployment Guide (v1.2)

This guide provides the complete, gap-free procedure for deploying the EduTracker Enterprise ERP at a client's institution. It replaces all previous PDF manuals and includes critical configuration steps for environment variables and network security.

Phase 1: Preparing the Distribution (On Your Machine)

Before visiting the client, you must create a secure, production-ready package. Run this on your developer machine:

1. **Open PowerShell as Administrator.**
 2. **Navigate to Project Root:** `powershell cd D:\cli\eduTracker`
 3. **Run the Bundler Command:** `powershell powershell -ExecutionPolicy Bypass -File bundle_release.ps1`
 4. **Verify the Package:**
 - Locate the newly created folder: `EduTracker_Release`.
 - **CRITICAL:** Ensure there are **NO** `src` folders inside the package.
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Phase 2: Client Machine Prerequisites

Ensure the client's "Server PC" meets these requirements before starting the installation.

1. **Operating System:** Windows 10/11 or Windows Server.
 2. **Node.js (LTS):** Download and install the latest LTS version from nodejs.org.
 3. **PostgreSQL (v16+):**
 - Install PostgreSQL.
 - Set the `postgres` superuser password to: `123456`.
 - Create an empty database named: `edutracker`.
 4. **Static IP (Highly Recommended):**
 - Assign a static IP to the Server PC (e.g., `192.168.0.102`) via Windows Network Settings to prevent the system from breaking after a router reboot.
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Phase 3: Installation & Configuration (At Client Office)

3.1 Transfer & Filesystem

1. **Copy Folder:** Paste the `EduTracker_Release` folder to the client's C: or D: drive.
2. **Permissions:** Right-click the folder > Properties > Security > Ensure 'Users' have Read/Write permissions.

3.2 Environment Configuration (Manual Step)

You must configure the backend to talk to the database.

1. Navigate to `EduTracker_Release/backend/`.
2. Create or edit a file named `.env` and paste the following:

```
# Database Connection
DATABASE_URL="postgres://postgres:123456@localhost:5432/edutracker?schema=public"

# Server Config
PORT=5000
NODE_ENV=production

# Security (Generate a unique string for each client)
JWT_SECRET="ET_ERP_$(Get-Random -Minimum 100000 -Maximum 999999)"
```

3.3 Automated Setup

1. **Open PowerShell as Administrator.**
2. Navigate to the `EduTracker_Release` folder.
3. **Enable Scripts:** `powershell Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser`
4. **Run Installer:** `powershell ./setup.ps1`
 - **What this does:** Detects the local IP, configures the frontend `.env`, installs dependencies, generates the database schema, and creates a Desktop shortcut.

Phase 4: Network & Firewall (Crucial Gap)

By default, Windows Firewall will block other PCs from accessing the ERP. You **MUST** perform these steps:

1. Open **Windows Defender Firewall with Advanced Security**.
2. Click **Inbound Rules > New Rule**.
3. Select **Port** > Next.
4. Select **TCP** and enter: 5000, 6001.
5. Select **Allow the connection** > Next.

6. Ensure Domain, Private, and Public are checked.
 7. Name the rule: **EduTracker ERP Ports**.
 8. Finish.
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Phase 5: Launching & Access

1. **Start the Engine:** Double-click `START_SYSTEM.bat`.
 - *Note: This terminal window must stay open for the system to work.*
 2. **Access the ERP:**
 - **On the Server:** Double-click the “EduTracker ERP” desktop short-cut.
 - **On Other Devices:** Open a browser and type: `http://[SERVER_IP]:6001` (e.g., `http://192.168.0.102:6001`).
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Phase 6: Maintenance

1. **Backups:** Train the client to go to **Settings > Database & Backup** weekly to click “Backup Now”.
 2. **Updates:** To update, replace the `backend/dist` and `.next` folders with new versions from your dev machine and restart `START_SYSTEM.bat`.
 3. **Troubleshooting Images:** If images don’t load, ensure `next.config.ts` has `unoptimized: true` (this is already set in the current version).
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